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1 $\forall X_0 \in \mathbb{R}^n \ \forall \delta > 0$

$$\begin{aligned} \text{Int } B(X_0\delta) &= B(X_0\delta) \\ \text{Ext } B(X_0\delta) &= \{X \in \mathbb{R}^n \mid \|X - X_0\| > \delta\} \\ \partial B(X_0\delta) &= \{X \in \mathbb{R}^n \mid \|X - X_0\| = \delta\} \\ \overline{B(X_0\delta)} &= \{X \in \mathbb{R}^n \mid \|X - X_0\| \leq \delta\} \end{aligned}$$

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- $\cdot : \mathbb{R}^n$ Cauchy
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- $\vdots$
- $\vdots \mathbb{R}^n$
- $\vdots \mathbb{R}^n$

$$\begin{array}{l}
 \S 2 \text{ } n \text{ } n \\
 \textcolor{red}{:} \text{ } XY \\
 \textcolor{blue}{1} \text{ } mn \geq 1 \text{ } \Omega \subseteq \mathbb{R}^n \\
 \textcolor{blue}{f} \text{ } \Omega \rightarrow \mathbb{R}^m \text{ } \Omega \text{ } n \\
 \text{ } m = 1 \text{ } n \text{ } (\text{ })
 \end{array}
 \begin{array}{l}
 x \in X \text{ } Y \\
 y(\text{ } y = f(x)) \\
 f \text{ } X \text{ } Y \text{ } (\text{ }) \text{ } y \text{ } x \text{ } x \text{ } y \\
 f \text{ } \textcolor{blue}{:} \text{ } X \rightarrow Y
 \end{array}$$

- $\quad \quad \quad \mathbf{f}\mathbf{g} : \Omega \rightarrow \mathbb{R}^m \quad \quad \lambda\mu \in \mathbb{R} \ \forall X \in \Omega$
 $(\lambda\mathbf{f} + \mu\mathbf{g})(X) = \lambda\mathbf{f}(X) + \mu\mathbf{g}(X)$
- $\quad \quad \quad \mathbf{f} : \Omega \rightarrow \mathbb{R}^m \quad \quad g : \Omega \rightarrow \mathbb{R} \quad \forall X \in \Omega \quad (g\mathbf{f})(X) := g(X)\mathbf{f}(X)$
 $(\frac{\mathbf{f}}{g})(X) := \frac{\mathbf{f}(X)}{g(X)} \ (g(X) \neq 0)$

- : $lmn \geq 1 \quad \Omega_1 \subseteq \mathbb{R}^n$
 $\Omega_2 \subseteq \mathbb{R}^m \quad \mathbf{f}: \Omega_1 \rightarrow \Omega_2 \quad \mathbf{g}: \Omega_2 \rightarrow \mathbb{R}^l$
 $\forall X \in \Omega_1 \quad (\mathbf{g} \circ \mathbf{f})(X) := \mathbf{g}(\mathbf{f}(X))$
- : $\mathbf{f}: \Omega \rightarrow \mathbb{R}^{m \times n} \quad \forall X = (x_1 \dots x_n) \in \Omega \quad \mathbf{f}(X) \in \mathbb{R}^m \quad Y = (y_1 \dots y_m)^T \quad y_j(X) : y_j = f_j(X) = f_j(x_1 \dots x_n) \quad \mathbf{f}: \Omega \rightarrow \mathbb{R}^{m \times n} \quad f_j: \Omega \rightarrow \mathbb{R} \quad \mathbf{f} = (f_1 \dots f_m)^T$

§3

$$\begin{aligned} & \text{1} \quad mn \geq 1 \quad \Omega \subset \mathbb{R}^n \\ & X_0 \in \mathbb{R}^n \quad \overline{\Omega} \quad \mathbf{f}: \overline{\Omega} \rightarrow \mathbb{R}^m \quad A \in \mathbb{R}^m \quad \forall \varepsilon > 0 \quad \exists \delta > 0 \quad \forall X \in \Omega \\ & 0 < \|X - X_0\|_n < \delta \quad \|\mathbf{f}(X) - A\|_m < \varepsilon \\ & X \in \Omega \quad X_0 \in \overline{\Omega} \quad \mathbf{f}(X_0) = A \\ & (\quad A) \quad \lim_{\Omega \ni X \rightarrow X_0} \mathbf{f}(X) = A \end{aligned}$$

- $\lim_{\Omega \ni X \rightarrow X_0} \mathbf{f}(X) = A \quad \forall \varepsilon > 0 \quad \exists \delta > 0 \quad \forall X \in \overset{\circ}{B}(X_0, \delta) \cap \Omega \quad \mathbf{f}(X)(A, \varepsilon)$
- $\mathbf{f} = (f_1 \dots f_m)^T \quad A = (a_1 \dots a_m)^T \quad \lim_{\Omega \ni X \rightarrow X_0} \mathbf{f}(X) = A \quad 1 \leq j \leq m \quad \lim_{\Omega \ni X \rightarrow X_0} f_j(X) = a_j$
- $X_0 \in \Omega \cup \{X_0\}$
 $\lim_{\Omega \ni X \rightarrow X_0} \mathbf{f}(X) \quad \lim_{X \rightarrow X_0} \mathbf{f}(X)$

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- : $fg : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{R} \quad \lambda \mu \in \mathbb{R} \quad \lim_{\Omega \ni X \rightarrow X_0} f(X) = A \quad \lim_{\Omega \ni X \rightarrow X_0} g(X) = B$
 - (a) $\lim_{\Omega \ni X \rightarrow X_0} (\lambda f + \mu g)(X) = \lambda A + \mu B$
 - (b) $\lim_{\Omega \ni X \rightarrow X_0} (fg)(X) = AB$
 - (c) $\lim_{\Omega \ni X \rightarrow X_0} \frac{f}{g}(X) = \frac{A}{B} \quad (B \neq 0)$

$$\begin{aligned}
\bullet \quad & \text{ : } \quad lmn \geq 1 \quad \Omega_1 \subseteq \mathbb{R}^n \quad \Omega_2 \subseteq \mathbb{R}^m \quad \boldsymbol{f} \colon \Omega_1 \rightarrow \Omega_2 \quad \boldsymbol{g} \colon \Omega_2 \rightarrow \mathbb{R}^l \\
& \quad \quad \quad \lim_{\Omega_1 \ni X \rightarrow X_0} \boldsymbol{f}(X) = Y_0 \quad \boldsymbol{f} \colon X_0 \rightarrow Y_0 \quad \lim_{\Omega_2 \ni Y \rightarrow Y_0} \boldsymbol{g}(Y) = A \\
& \quad \quad \quad \lim_{\Omega_1 \ni X \rightarrow X_0} (\boldsymbol{g} \circ \boldsymbol{f})(X) = A
\end{aligned}$$

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- **Sequential** : $m, n \geq 1 \quad \Omega \subseteq \mathbb{R}^n \quad \mathbf{f}: \Omega \rightarrow \mathbb{R}^m \quad X_0 \in \mathbb{R}^n \quad A \in \mathbb{R}^m \quad \lim_{\Omega \ni X \rightarrow X_0} \mathbf{f}(X) = A \quad \Omega \setminus \{X_0\} \quad X_0 \quad \{X_k\} \quad \lim_{k \rightarrow \infty} \mathbf{f}(X_k) = A$
- **Cauchy** : $\lim_{\Omega \ni X \rightarrow X_0} \mathbf{f}(X)$
 $\forall \varepsilon > 0 \exists \delta > 0 \quad \forall X' X'' \in \mathring{B}(X_0, \delta) \cap \Omega$
 $\|\mathbf{f}(X') - \mathbf{f}(X'')\|_m < \varepsilon$

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow \infty}} f(xy)$$

$$\lim_{\substack{x \rightarrow \infty \\ y \rightarrow \infty}} f(xy)$$

$$\lim_{\substack{x \rightarrow \infty \\ y \rightarrow +\infty}} f(xy)$$

$$\lim_{\substack{(xy) \rightarrow (x_0 y_0)}} f(xy)$$

$$\lim_{\substack{x \rightarrow x_0 \\ y \rightarrow y_0}} f(xy)$$

$$x_0 \quad y_0$$

3 $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow a}} \left(1 - \frac{1}{2x}\right)^{\frac{x^2}{x+y}} \quad a \in \mathbb{R}$

$$\lim_{\substack{x \rightarrow \infty \\ y \rightarrow a}} \left(1 - \frac{1}{2x}\right)^{\frac{x^2}{x+y}} = e^{-\frac{1}{2}}$$

$$\begin{aligned} &: \lim_{\substack{x \rightarrow \infty \\ y \rightarrow a}} \ln \left(1 - \frac{1}{2x}\right)^{\frac{x^2}{x+y}} = \lim_{\substack{x \rightarrow \infty \\ y \rightarrow a}} \frac{x^2}{x+y} \ln \left(1 - \frac{1}{2x}\right) \\ &= \lim_{\substack{x \rightarrow \infty \\ y \rightarrow a}} \frac{x^2}{x+y} \cdot \frac{-1}{2x} = \lim_{\substack{x \rightarrow \infty \\ y \rightarrow a}} \frac{-1}{2} \cdot \frac{1}{1 + \frac{y}{x}} = -\frac{1}{2} \end{aligned}$$

$$\begin{array}{l}
4 \qquad \lim_{(xy) \rightarrow (00)} \frac{2xy}{x^2+y^2} \\
: \forall (xy) \in \mathbb{R}^2 \setminus \{(00)\} \quad g(xy) = \frac{2xy}{x^2+y^2} \qquad A \, \forall kx \in \mathbb{R} \quad f_k(x) = (xkx) \quad \lim_{x \rightarrow 0} f_k(x) = (00) \quad f_k \, \mathbb{R} \setminus \{0\} \quad (00) \qquad A = \lim_{x \rightarrow 0} g(xkx) = \frac{2k}{1+k^2} \qquad !
\end{array}$$

$$5 \quad f(xy) = \frac{x^2}{x^2+y^2-x} (xy) \qquad (00) \qquad 0 \quad (xy) \quad (00)$$

$$: \quad ab \in \mathbb{R} \qquad (00) \qquad \begin{cases} x = at \\ y = bt \end{cases} (t \in \mathbb{R})$$

$$\lim_{t \rightarrow 0} f(atbt) = \lim_{t \rightarrow 0} \frac{(at)^2}{(at)^2 + (bt)^2 - at} = \lim_{t \rightarrow 0} \frac{a^2}{a^2 + b^2 - \frac{a}{t}} = 0$$

$$\forall t \in \mathbb{R} \quad g(t) = (t^2t) \quad \lim_{t \rightarrow 0} g(t) = (00) \quad g \, \mathbb{R} \setminus \{0\} \quad (00)$$

$$\lim_{t \rightarrow 0} f \circ g(t) = \lim_{t \rightarrow 0} \frac{(t^2)^2}{(t^2)^2 + t^2 - t^2} = 1 \neq 0$$

$$\lim_{(xy) \rightarrow (00)} f(xy)$$

6 $\lim_{(xy) \rightarrow (00)} \frac{\sin(x^2 + e^y)}{1 + \ln(1+x)}$

: $\lim_{(xy) \rightarrow (00)} \frac{\sin(x^2 + e^y)}{1 + \ln(1+x)} = \sin 1$

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$$\lim_{(xy) \rightarrow (x_0 y_0)} f(xy)$$

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$$\lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(xy) \quad \lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(xy)$$

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$$\lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(xy) \quad x \neq x_0 \quad \varphi(x) := \lim_{y \rightarrow y_0} f(xy) \quad \lim_{x \rightarrow x_0} \varphi(x)$$

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$$\lim_{(xy) \rightarrow (00)} \frac{2xy}{x^2+y^2} \quad y \neq 0 \quad \lim_{x \rightarrow 0} \frac{2xy}{x^2+y^2} = 0$$

$$\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} \frac{2xy}{x^2+y^2} = 0$$

$$\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} \frac{2xy}{x^2+y^2} = 0$$

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8 $\forall (xy) \in \mathbb{R}^2 (\ xy \neq 0)$

$$\lim_{x \rightarrow 0} x \sin \frac{1}{y} = 0$$

$$\lim_{x \rightarrow 0} y \sin \frac{1}{x}$$

$$\lim_{x \rightarrow 0} f(xy)$$

$$\lim_{y \rightarrow 0} f(xy)$$

$$|f(xy)| \leq |x| + |y|$$

$$\lim_{(xy) \rightarrow (00)} f(xy) = 0$$

$$f(xy) = x \sin \frac{1}{y} + y \sin \frac{1}{x}$$

$$\lim_{(xy)\rightarrow(x_0y_0)}f(xy)=A \quad x_0 \in U \quad \lim_{y\rightarrow y_0}f(xy)=\varphi(x)$$

$$A=\lim_{x\rightarrow x_0}\varphi(x)=\lim_{x\rightarrow x_0}\lim_{y\rightarrow y_0}f(xy)$$

$$\forall \varepsilon>0 \quad \exists \delta>0 \quad \forall (xy)\in \overset{\circ}{B}((x_0y_0)\delta) \quad |f(xy)-A|<\varepsilon \quad \forall x\in U\cap (x_0-\delta x_0+\delta) \quad y \quad |\varphi(x)-A|\leq \varepsilon \quad \lim_{x\rightarrow x_0}\varphi(x)=A$$

$$A\in \mathbb{R}$$

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$$\lim_{(xy) \rightarrow (x_0 y_0)} f(xy) = A$$

$$\lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(xy) = B \quad A = B$$

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$$\lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(xy) \quad \lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(xy)$$

$$\lim_{(xy) \rightarrow (x_0 y_0)} f(xy)$$

- $$\begin{array}{llll} \bullet & : & n & n \left(\begin{array}{l} \\ \end{array} \right) \\ \bullet & : & & ; \quad ; \\ \bullet & : & \mathbb{R}^m & n \\ & & m & n \end{array}$$



- $\lim_{\Omega \ni X \rightarrow X_0} \mathbf{f}(X)$ $\lim_{X \rightarrow X_0} \mathbf{f}(X)$
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- Cauchy



- $\lim_{(xy) \rightarrow (00)} \frac{\sin(x^2+y^2)}{x^2+y^2} = 1$
- $\lim_{(xy) \rightarrow (00)} \frac{xy}{\sqrt{x^2+y^2}} = 0$
- $\lim_{\substack{x \rightarrow \infty \\ y \rightarrow a}} \left(1 - \frac{1}{2x}\right)^{\frac{x^2}{x+y}} = e^{-\frac{1}{2}} \quad a \in \mathbb{R}$
- $\lim_{(xy) \rightarrow (00)} \frac{2xy}{x^2+y^2}$

:
$$\lim_{(xy) \rightarrow (x_0y_0)} f(xy)$$

:
$$\lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(xy) \quad \lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(xy)$$

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$$\begin{array}{l} 2 \\ \Omega \end{array} \quad mn \geq 1 \quad \Omega \subseteq \mathbb{R}^n \quad X_0 \in \Omega \\ \mathbf{f}: \Omega \rightarrow \mathbb{R}^m$$

$$\lim_{\Omega \ni X \rightarrow X_0} \mathbf{f}(X) = \mathbf{f}(X_0)$$

$$\mathbf{f}(X_0)$$

- $\mathbf{f} : X_0 \rightarrow \mathbb{R}^m$
 $\forall \varepsilon > 0 \exists \delta > 0$
 $\forall X \in \Omega \quad \|X - X_0\|_n < \delta$

$$\|\mathbf{f}(X) - \mathbf{f}(X_0)\|_m < \varepsilon$$

- $\mathbf{f} : \Omega \rightarrow \mathbb{R}^m$
- $\mathcal{C}(\Omega; \mathbb{R}^m) = \{ \mathbf{f} \mid \mathbf{f} : \Omega \rightarrow \mathbb{R}^m \}$
- $m = 1$

$$\mathcal{C}(\Omega)$$

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

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$$\begin{aligned}
&\Omega \subset \mathbb{R}^n \quad \mathbf{f} \colon \Omega \rightarrow \mathbb{R}^m \quad \mathbf{f} \colon \mathbb{R}^m \rightarrow G \quad \mathbf{f}^{-1}(G) = \{x \in \Omega \mid \mathbf{f}(x) \in G\} \\
&\mathbb{R}^m \quad G \quad \mathbf{f}^{-1}(G) \quad X_0 \in \Omega \quad \forall \varepsilon > 0 \quad G = B(\mathbf{f}(X_0)\varepsilon) \quad \mathbf{f}^{-1}(G) \quad X_0 \quad \exists \delta > 0 \quad B(X_0\delta) \subseteq \mathbf{f}^{-1}(G) \quad \forall X \in B(X_0\delta) \quad \|\mathbf{f}(X) - \mathbf{f}(X_0)\|_m < \varepsilon \quad \mathbf{f} \colon X_0 \rightarrow \mathbf{f}
\end{aligned}$$

$$\mathbf{f} : G \rightarrow \mathbb{R}^m \quad \forall X_0 \in \mathbf{f}^{-1}(G) \quad \mathbf{f}(X_0) \in G \quad G \quad \exists \varepsilon > 0 \quad B(\mathbf{f}(X_0), \varepsilon) \subseteq G \quad \mathbf{f}(X_0) \quad \exists \delta_1 > 0 \quad \forall X \in \Omega \cap B(X_0, \delta_1) \quad \|\mathbf{f}(X) - \mathbf{f}(X_0)\|_m < \varepsilon \quad \Omega \cap B(X_0, \delta_1) \quad \exists \delta > 0 \quad B(X_0, \delta) \subseteq \Omega \cap B(X_0, \delta_1) \quad \forall X \in B(X_0, \delta) \quad \|\mathbf{f}(X) - \mathbf{f}(X_0)\|_m < \varepsilon$$

$$B(X_0\delta) \subseteq \boldsymbol{f}^{-1}\big(B(\boldsymbol{f}(X_0)\varepsilon)\big) \subseteq \boldsymbol{f}^{-1}(G) \quad X_0 \in \boldsymbol{f}^{-1}(G) \quad \boldsymbol{f}^{-1}(G)$$

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∴ $\boldsymbol{f} : \mathbb{R}^m \rightarrow G \subseteq \boldsymbol{f}^{-1}(G)$

5 () $\Omega \subseteq \mathbb{R}^n$ $f \in \mathcal{C}(\Omega)$ $f|_{\Omega}$
: $f|_{\Omega}$ $\forall k \in \mathbb{N}^* \exists X_k \in \Omega |f(X_k)| > k$ Ω $\{X_k\}$ $\{X_{\ell_k}\}$ $A \subset \Omega$ $A \in \Omega$ f $f(A) = \lim_{k \rightarrow \infty} f(X_{\ell_k}) = \infty$! f



$$mn \geq 1 \quad \Omega \subseteq \mathbb{R}^n \quad X_0 \in \Omega \quad \Omega \quad \mathbf{f}: \Omega \rightarrow \mathbb{R}^m$$

$$\lim_{\Omega \ni X \rightarrow X_0} \mathbf{f}(X) = \mathbf{f}(X_0)$$

$$\mathbf{f}(X_0)$$

$$\Omega \subset \mathbb{R}^n \quad \mathbf{f}: \Omega \rightarrow \mathbb{R}^m \quad \mathbf{f}: \mathbb{R}^m \rightarrow G \quad \mathbf{f}^{-1}(G) = \{x \in \Omega \mid \mathbf{f}(x) \in G\}$$

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$$(\quad) \quad \Omega \subseteq \mathbb{R}^n \quad f \in \mathcal{C}(\Omega) \quad f: \Omega \rightarrow \mathbb{R}^m$$

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$$D \subseteq \mathbb{R}^n \quad \forall X, Y \in D \quad D \quad XY \quad \gamma : [0, 1] \rightarrow D \quad \gamma(0) = X \quad \gamma(1) = Y$$

$$\gamma : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

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$f \in \mathcal{C}(\Omega; \mathbb{R}^m)$

$\Omega \subseteq \mathbb{R}^n$

$f(\Omega)$

$\forall X_1 X_2 \in \Omega$

$f(X_1) \neq f(X_2)$

$\mu \int_{\Omega} f(X_0) d\mu = \int_{\Omega} f(X_0) d\mu$

$$f : \Omega \rightarrow \mathbb{R} \quad \forall X_1, X_2 \in \Omega \quad f(X_1) \leq f(X_2) \iff f(X_1) = f(X_2)$$

$$\overset{\textcolor{blue}{1}}{X} = \overset{mM}{(x_1 \dots x_n)} \in \mathbb{R}^n$$

$$\textcolor{red}{\text{Analysis:}} \quad X \qquad X \qquad \frac{1}{M} \leq \sum_{j=1}^n \left| \frac{x_j}{\|X\|} \right| \leq \frac{1}{m} \quad y_j = \frac{x_j}{\|X\|} \quad Y = (y_1 \dots y_2) \quad \|Y\| = 1$$

$$\textcolor{red}{m} \sum_{j=1}^n |x_j| \leq \|X\| \leq M \sum_{j=1}^n |x_j|$$

$$f: S = \{Y \in \mathbb{R}^n \mid \|Y\|_n = 1\} \rightarrow S \quad \forall Y = (y_1 \dots y_n) \in S$$

$$f(a) > 0 \quad b$$

$$\frac{1}{M} \leq f(Y) := \sum_{j=1}^n |y_j| \leq \frac{1}{m}$$

$$f(Y) = \sum_{j=1}^n |y_j| > 0$$

$$m = \frac{1}{b} \quad M = \frac{1}{a} \quad \forall X \in \mathbb{R}^n (X \neq 0) \quad Y = \frac{1}{\|X\|_n} (x_1 \dots x_n) \in S \quad a \leq f(Y) \leq b \quad a \leq \frac{1}{\|X\|_n} \sum_{j=1}^n |x_j| \leq b$$

$$m \sum_{j=1}^n |x_j| \leq \|X\|_n \leq M \sum_{j=1}^n |x_j| \quad X \in \mathbb{R}^n$$

$$\frac{1}{b} \sum_{j=1}^n |x_j| \leq \|X\|_n \leq \frac{1}{a} \sum_{j=1}^n |x_j|$$

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$n \geq 1 \quad \Omega \subseteq \mathbb{R}^n \quad X_0 \in \mathbb{R}^n \quad \Omega \quad f: \Omega \rightarrow \mathbb{R}$

(1)

$\lim_{\Omega \ni X \rightarrow X_0} f(X) = 0 \quad f \quad \Omega \ni X \rightarrow X_0$

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$\lim_{\Omega \ni X \rightarrow X_0} f(X) = A$

$f(X) = o(1) \quad (\Omega \ni X \rightarrow X_0)$

$f(X) - A = o(1) \quad (\Omega \ni X \rightarrow X_0)$

$$(2) \quad g : \Omega \rightarrow \mathbb{R} \qquad \beta > 0 \; \delta > 0 \; \forall X \in \Omega \cap \mathring{B}(X_0 \delta) \quad |f(X)| \leq \beta |g(X)|$$

$$(3) \quad \begin{array}{l} g(X) = O(f(X)) \quad fg \\ k \geq 0 \quad \lim_{\Omega \ni X \rightarrow X_0} \frac{f(X)}{\|X - X_0\|^k} = 0 \quad f \quad \Omega \ni X \rightarrow X_0 \quad \|X - X_0\|^k \\ f(X) = o(\|X - X_0\|^k) \quad (\Omega \ni X \rightarrow X_0) \end{array}$$

$$f(X) = O(g(X)) \; (\Omega \ni X \rightarrow X_0)$$

$$(4) \quad \lim_{\Omega \ni X \rightarrow X_0} \frac{f(X)}{\|X - X_0\|^k} = c \neq 0 \quad \text{if } \Omega \ni X \rightarrow X_0 \quad \|X - X_0\| \leq k \quad \text{if } k = 0 \quad \lim_{\Omega \ni X \rightarrow X_0} f(X) = c \quad 0$$

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